ORF307: Optimization

Precept: #11

Agenda

- Branch & Bound overview
- Disjunctive constraints problem
- Production/Distribution problem

Reminders and Announcements

- Always check Python notebooks in Companion (in directories `lectures/` and `precepts/`)
- Write down the deadline of the current HW: ....
- Other announcements on blackboard.

Mixed integer program

A general mixed integer programming (MIP) problem is defined as

$$(MIP) := \begin{cases} \min\limits_{x} \quad c^T x \\ s.t. \quad Ax \leq b \\ \qquad x_i \in \mathbb{Z}, \quad \forall i \in \mathcal{I} \\ \qquad x_j \in \mathbb{R}, \quad \forall j \in \{1, 2, \ldots, n\} \setminus \mathcal{I} \end{cases}$$

where $c, x \in \mathbb{R}^n, A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m$ and the set $\mathcal{I}$ contains the indices of the variables that must take integer values. Note that the symbol " $\setminus$ " represents the Set Difference operation[1]. Obviously $\mathcal{I} \subseteq \{1, 2, \ldots, n\}$.

An example of the feasible region of a MIP problem is shown in Figure 1. The feasible region is defined as

$$P^{(MIP)} = \{x \in \mathbb{R} \times \mathbb{Z} \mid x_1 + x_2 \leq 6, x_1 \leq 4.5, x_2 \leq 3.5, x_1 \geq 0, x_2 \geq 0\}$$

and consists of the points on the blue lines, where the variable $x_1 \in \mathbb{R}$, whereas the variable $x_2 \in \mathbb{Z}$. Note that in this example the set if integer variable indices is $\mathcal{I} = \{2\}$.

[1]The difference of two sets G and H is defined as the set $G \setminus H = \{x \mid x \in G \text{ and } x \notin H\}$

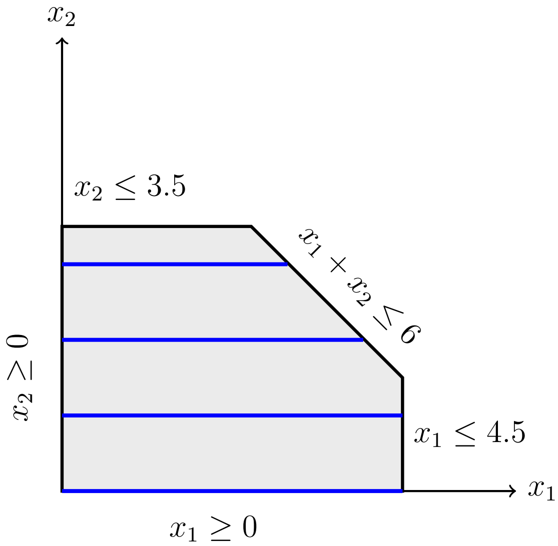

Figure 1: The feasible region is defined by the points on the blue lines (note that there are 4 blue lines)

If $\mathcal{I} = \{1, 2, \ldots, n\}$ then the problem is a (pure) integer programming (IP) problem and all variables must take integer values. Such a model is defined as

$$(IP) := \begin{cases} \min\limits_{x} \quad c^T x \\ s.t. \quad Ax \leq b \\ \qquad\ \ x \in \mathbb{Z}^n \end{cases}$$

Note that we have omitted the set $\mathcal{I}$ since all variables must be integer in this case. Figure 2 shows a modification of the previous 2-dimensional example where both x_1 and x_2 must take integer values. The feasible region of this problem is now defined as

$$P^{(IP)} = \{x \in \mathbb{Z}^2 | x_1 + x_2 \leq 6, x_1 \leq 4.5, x_2 \leq 3.5, x_1 \geq 0, x_2 \geq 0\}$$

Continuous Relaxations

The continuous relaxation of a MIP or IP problem is an LP problem that results from allowing the integer variables to take real values. Oftentimes the continuous relaxation is also referred to as LP relaxation[2]. The continuous relaxation of the (MIP) and (IP) problems we defined above is the LP problem:

$$(LP_{rel}) := \begin{cases} \min\limits_{x} \quad c^T x \\ s.t. \quad Ax \leq b \end{cases}$$

and its feasible region is the polyhedron

$$P^{(rel)} = \{x \in \mathbb{R}^n | Ax \leq b\}$$

[2]We will use both terms interchangeably.

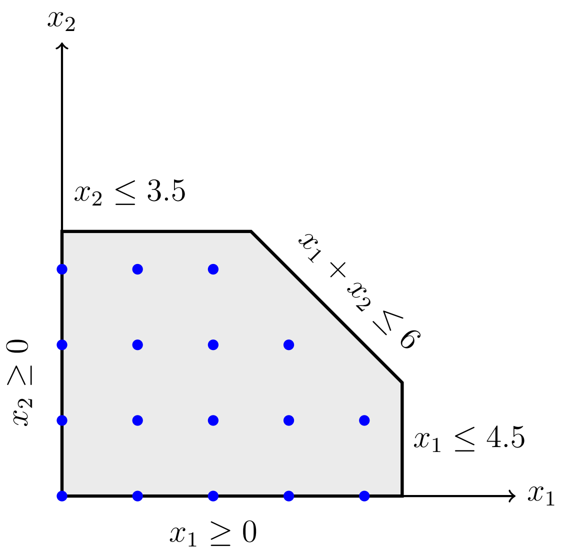

Figure 2: The feasible region of a pure IP. All variables must take integer values only. The feasible region consists of the blue points only.

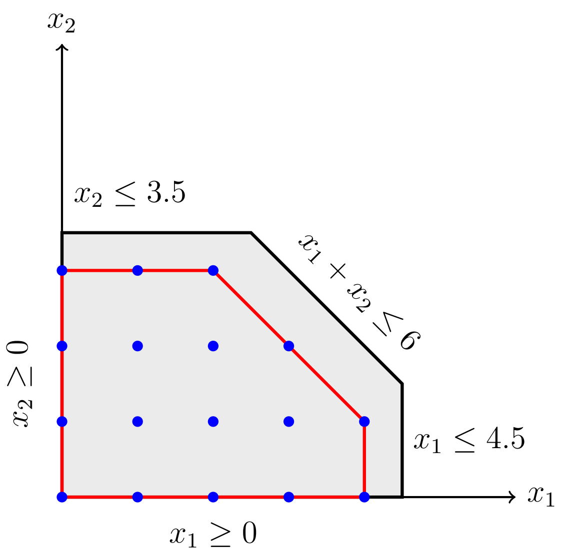

Figure 3: The convex hull of the feasible region of a pure IP is refined by the continuous space which is the intersection of the red hyperplanes. An LP whose feasible region is defined by the convex hull has the same optimal solution as the corresponding IP or MIP defined over the integer points within the convex hull (note that: the LP and the IP or MIP must have the same objective function).

The shaded (gray) region in Figures 1 and 2 show the feasible region of the LP relaxation of the MIP and the IP, which is the same polyhedron in both cases.

It is easy to see that the following relations hold:

$$P^{(IP)} \subset P^{(MIP)} \subset P^{(rel)} \tag{1}$$

From these relations we can derive the following relations for the **optimal objective**

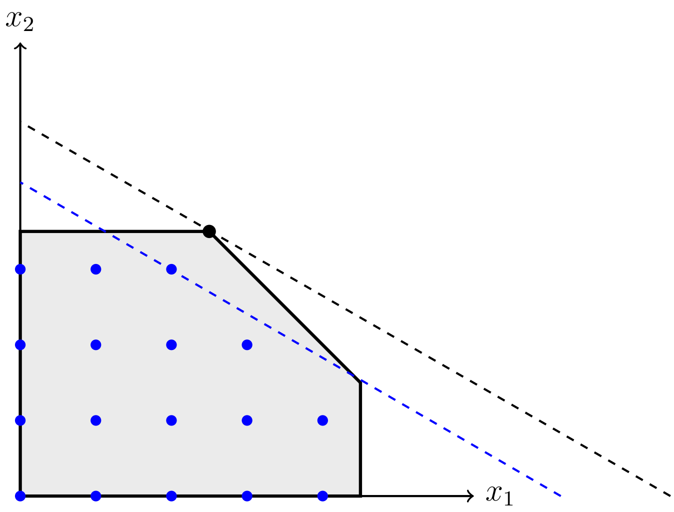

Figure 4: The objective function is shown in dashed lines. The optimal solution of the IP problem is $x^*_{IP} = (2, 3)$, whereas the optimal solution of the LP relaxation is $x^*_{LP} = (2.5, 3.5)$ (shown as the black dot) and is *not integer*.

values of the IP ($p^{*(IP)}$), the MIP ($p^{*(MIP)}$) and the LP relaxation ($p^{*(rel)}$):

$$p^{*(rel)} \leq p^{*(MIP)} \leq p^{*(IP)} \tag{2}$$

This is convenient way to obtain lower bounds of the MIP or IP problems. LP problems are easier to solve computationally using the Simplex method or Interior Point Methods.

Note (it is important to remember): if $x^{*(rel)} \in \mathbb{Z}^n$ then $x^{*(rel)} = x^{*(IP)}$. This happens when the convex hull of $P^{(IP)}$ is equal to $P^{(rel)}$ (see Figure 3 for an example).

Branch and Bound (B&B) method: a simple example

Let us first illustrate the main ideas of the B&B method by solving a simple IP problem:

$$IP \begin{cases} \min & -2x_1 - 3x_2 \\ \text{s.t.} & \dfrac{2}{9}x_1 + \dfrac{1}{4}x_2 \leq 1 \\ & \dfrac{1}{7}x_1 + \dfrac{1}{3}x_2 \leq 1 \\ & x \geq 0, \ \ x \in \mathbb{Z}^2 \end{cases} \tag{3}$$

Using this simple IP problem we will show how the B&B method progressively builds a binary tree (aka, B&B tree) and uses it to search for the optimal solution of the IP. The nodes of the tree are LP problems whose feasible region is a partition of the LP relaxation of the original IP problem. The edges represent additional constraints that are added to the subproblems.

The plot of the feasible region of this IP is shown in Figure 5 and is represented by the black points only. The integer optimal solution of this problem is the point $x^*_{IP} = (2, 2)$.

The green shaded area represents the feasible region of the continuous relaxation of the IP problem, and is an LP problem.

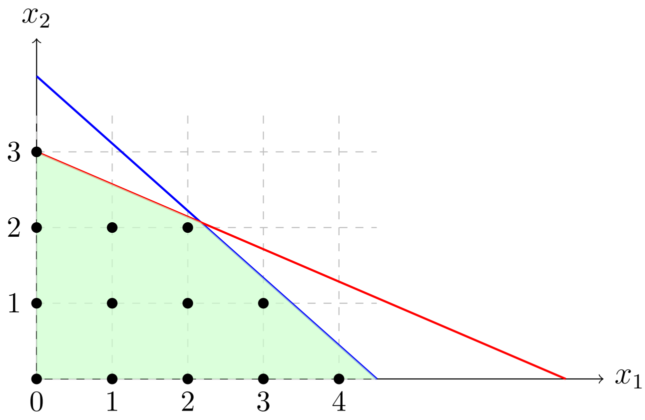

Figure 5: Plot of the root node of the B&B tree. The blue line and the red line represent the 1st and 2nd constraints respectively. The black dots are the integer feasible points and the green area represents the feasible region of the LP relaxation of the IP.

We initialize the lower bound $LB = -\infty$ and the upper bound $UB = +\infty$ (optimal objective value of the IP problem must be between $-\infty$ and $+\infty$). We then solve the LP relaxation of the original IP, which is defined as

$$LP_0 \begin{cases} \min & -2x_1 - 3x_2 \\ \text{s.t.} & \frac{2}{9}x_1 + \frac{1}{4}x_2 \leq 1 \\ & \frac{1}{7}x_1 + \frac{1}{3}x_2 \leq 1 \\ & x_1, x_2 \geq 0. \end{cases} \tag{4}$$

The problem LP_0 becomes the first element in the list of unexplored (active) problems $\Lambda = \{LP_0\}$, and represents the root node of the B&B tree.

We pick the problem LP_0 from the list and solve it using CvxPy (or any other optimization solver). Its optimal solution is $x^*_{LP_0} = (2.172, 2.069)$. This point is at the intersection of the two inequality constraints (i.e., the blue and the red lines in Figure 5). The optimal objective value of the LP_0 is $p^*_{LP_0} = -10.55$, and hence we have found our first meaningful lower bound, and we replace: $LB = -10.55$.

The optimal solution of LP_0 is fractional (none of its elements in an integer number) and since we have $LB = -10.55 < UB = +\infty$, we will branch on a fractional value, say $x_1 = 2.172$. Two new subproblems are created:

$$LP_1 \begin{cases} \min & -2x_1 - 3x_2 \\ \text{s.t.} & \frac{2}{9}x_1 + \frac{1}{4}x_2 \leq 1 \\ & \frac{1}{7}x_1 + \frac{1}{3}x_2 \leq 1 \\ & x_1, x_2 \geq 0 \\ & x_1 \leq \lfloor 2.172 \rfloor = 2 \end{cases} \qquad LP_2 \begin{cases} \min & -2x_1 - 3x_2 \\ \text{s.t.} & \frac{2}{9}x_1 + \frac{1}{4}x_2 \leq 1 \\ & \frac{1}{7}x_1 + \frac{1}{3}x_2 \leq 1 \\ & x_1, x_2 \geq 0 \\ & x_1 \geq \lceil 2.172 \rceil = 3 \end{cases}$$

The plots of the feasible regions of the two new LPs are shown in Figures 6 and 7. We add both LPs to the list of unexplored (active) problems which now becomes: $\Lambda = \{LP_1, LP_2\}$. Note that LP_0 has been deleted from the list, since we have solved (explored) it.

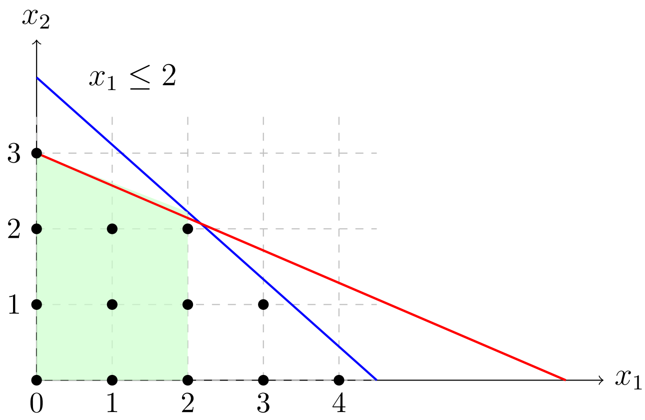

Figure 6: Plot of the feasible region of LP_1, the second node in the B&B tree.

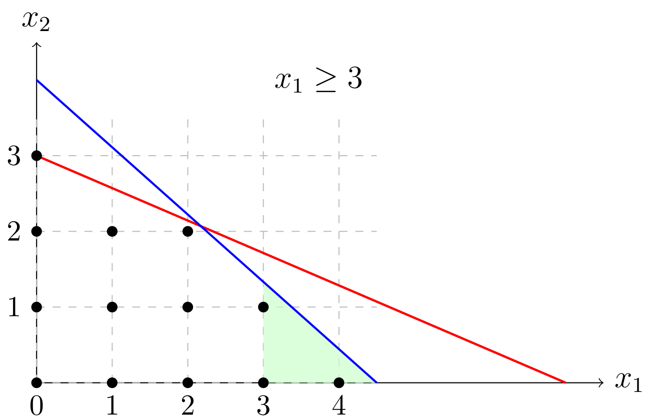

Figure 7: Plot of the feasible region of LP_2, the third node in the B&B tree.

We now pick LP_1 from the list and solve it (this also means that we delete it from the list). We find that its optimal solution is $x^*_{LP_1} = (2, 1.143)$ and its optimal objective function value is $p^*_{LP_1} = -10.43$.

Note that the optimal solution of LP_1 still contains a fractional value for the integer variable x_2. We will branch on x_2 and generate two new subproblems:

$$LP_3 \begin{cases} \min & -2x_1 - 3x_2 \\ \text{s.t.} & \frac{2}{9}x_1 + \frac{1}{4}x_2 \leq 1 \\ & \frac{1}{7}x_1 + \frac{1}{3}x_2 \leq 1 \\ & x_1, x_2 \geq 0 \\ & x_1 \leq \lfloor 2.172 \rfloor = 2 \\ & x_2 \leq \lfloor 2.143 \rfloor = 2 \end{cases} \qquad LP_4 \begin{cases} \min & -2x_1 - 3x_2 \\ \text{s.t.} & \frac{2}{9}x_1 + \frac{1}{4}x_2 \leq 1 \\ & \frac{1}{7}x_1 + \frac{1}{3}x_2 \leq 1 \\ & x_1, x_2 \geq 0 \\ & x_1 \leq \lfloor 2.172 \rfloor = 2 \\ & x_2 \geq \lceil 2.143 \rceil = 3 \end{cases}$$

The plots of the feasible regions of the two new LPs are shown in Figures 8 and 9. We add the two new subproblems to the list of active problems, which now becomes: $\Lambda = \{LP_3, LP_4, LP_2\}$.

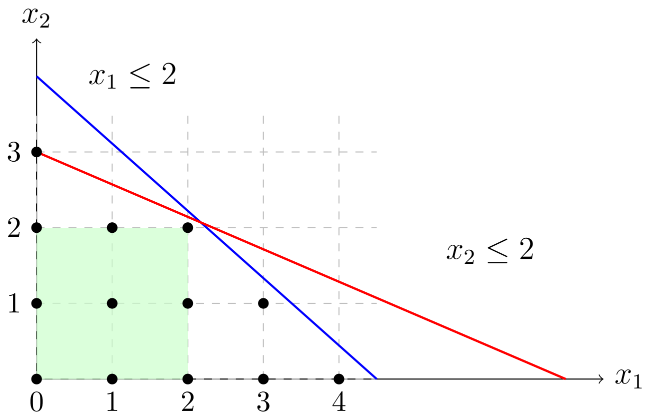

Figure 8: Plot of the feasible region of LP_3, the fourth node in the B&B tree. Note that the original linear constraints (the blue and red line segments) have become redundant, and the new constraints, $x_1 \leq 2$ and $x_2 \leq 2$ (together with the non-negativity constraints), define the feasible region of this subproblem.

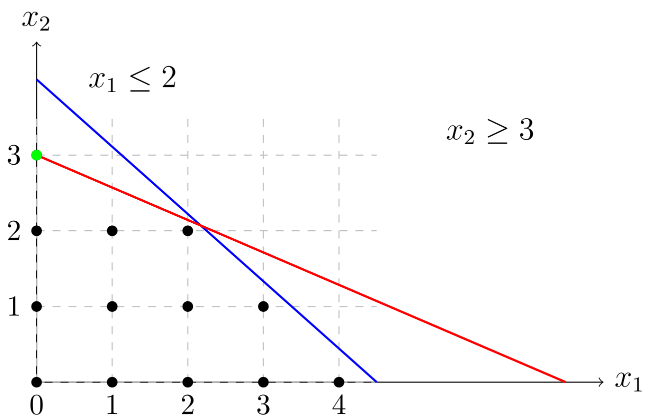

Figure 9: Plot of the feasible region of LP_4, the fifth node in the B&B tree. Note that this feasible region contains only one point $(0, 3)$. Note also that the blue constraint becomes redundant.

Next we solve LP_3 and delete from the list. We find that its solution is $x^*_{LP_3} = (2,2)$ and its optimal objective value is $p^*_{LP_3} = -10$. Since this is a larger value than the current lower bound, we update the lower bound to $LB = -10$. In addition, $x^*_{LP_3} = (2,2)$ is a feasible solution of the original IP problem, since both x_1 and x_2 have integer values. We call this solution the **incumbent solution**, as it represents the best solution of the IP problem we have found so far. We will denote it $x_{inc} = (2,2)$. This also means we have found an upper bound of the optimal solution of the original IP. We therefore update the upper bound to $UB = -10$ (form the initial value $UB = +\infty$). Also, we do not need to branch on any variable in this node, since they both have integer values. The list of active subproblem now has been reduced by one, $\Lambda = \{LP_4, LP_2\}$, and therefore we did not generate any new subproblems (the list of unexplored subproblems beocmes smaller).

We now solve LP_4 (using CvxPy) and find that its solution is $x^*_{LP_4} = (0,3)$. This is the only feasible point of the feasible region of LP_4. It also has integer values and it is therefore a feasible solution of the original IP. The optimal objective value of LP_4 is $p^*_{LP_4} = -9$, and is greater than the upper bound $UB = -10$. This means that this integer feasible solution is worse than (or inferior to) the incumbent solution $x_{inc} = (2,2)$, and therefore we ignore it. Note that if it happened to be better than the incumbent solution (i.e., its optimal objective was less than -10), we would have updated the value of x_{inc}, to keep track of the best integer solution we have found. Also, we do need to branch on any variable in this subproblem, since they both have integer values. Note that the list of active subproblem now has now been reduced by one, $\Lambda = \{LP_2\}$.

We finally pick LP_2 from the list and solve it. Its solution is: $x^*_{LP_2} = (3, 1.33)$, which is not integer, and its optimal objective value $p^*_{LP_2} = -10$. Note that the optimal objective value $p^*_{LP_2}$ is greater than or equal to the lower bound $UB = -10$ (that is, $UB = -10 \leq -10 = p^*_{LP_2}$). Hence, even if we branch on the fractional value 1.33 and create two additional LPs (LP_2 with $x_2 \leq \lfloor 1.33 \rfloor$ and LP_2 with $x_2 \geq \lceil 1.33 \rceil$) their objective values will be greater than $UB = -10$. We can therefore safely prune all its children nodes (i.e., we do not need to branch on any of its fractional values that correspond to integer variables), since they will not contain any better solution than the incumbent.

The list of active subproblems has now become empty ($\Lambda = \emptyset$), meaning that there are no more unexplored subproblems and therefore the B&B method has completed its journey. The optimal solution of the original IP problem is the incumbent solution, that is $x^*_{IP} = x_{inc} = (2,2)$. The entire B&B tree is shown in Figure 10.

Branch and Bound (B&B) method

Main steps (in words)

The core idea of the B&B method is to iteratively divide (branch) the original IP problem (or MIP problem) into smaller smaller ones and then using bounds (derived from the relations shown in (1) and (2)) to eliminate subproblems that cannot contain the integer optimal solution. The B&B methods can be thought of as a *tree search* where nodes are the subproblems and edges represent the branching constraints used to divide the feasible region.

The **main steps** of the B&B methods can be summarized as follows:

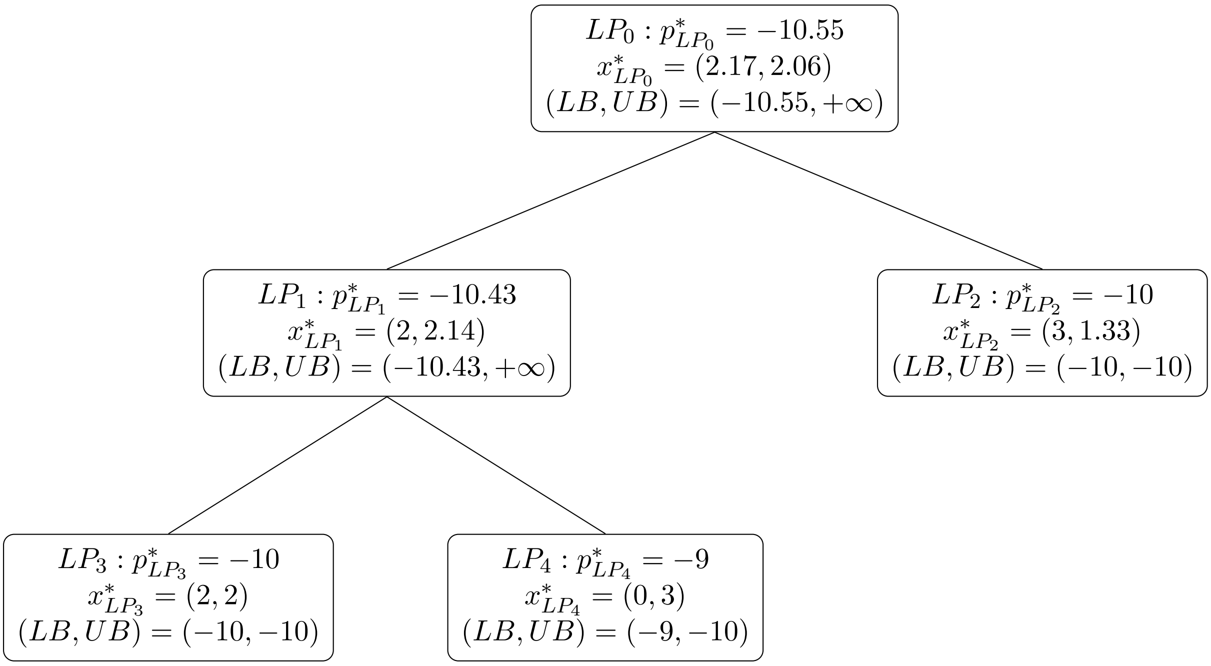

Figure 10: The B&B tree with the optimal solutions of the LP relaxations and the lower and upper bound updates. Note how the lower bound and upper bound are updated based on the optimal objective values of the LP relaxations, and the type (integer or fractional) of the LP optimal solution.

1. Start by solving the LP relaxation of the initial IP. This is the *root node* of the tree search.

   If the optimal solution of the LP relaxation (of the root node) satisfies all integer constraints, then we have found the optimal solution of the IP, and we *STOP*.

   Otherwise, the optimal objective value of the LP relaxation provides a lower bound (LB) on the optimal objective value of the original IP (this is guaranteed by (2)).

   In addition, an initial upper bound (UB) on the original IP can be set to $+\infty$, or an arbitrary feasible integer solution (usually computed by some heuristic).

2. Branching: Suppose x^* is the optimal solution of the LP-relaxation of the original IP and there are some integer variables that have fractional values. Choose a violated integer variable (i.e., one that has fractional value) and branch on it.

   More specifically, let's say x_i is an integer variable in the original MIP (i.e., $i \in \mathcal{I}$), but in the optimal solution of the current LP relaxation it has a fractional value (i.e., $x_i^* \in \mathbb{R}$).

   If $LB < UB$ the subproblem is divided into two new subproblems by adding the new constraint:

   - subproblem 1: add the constraint $x_i \leq \lfloor x_i^* \rfloor$ to the feasible region of the current subproblem.
   - subproblem 2: add the constraint $x_i \geq \lceil x_i^* \rceil$ to the feasible region of the current subproblem

Both subproblems become children of the current subproblem. We add them to **the list of problems to be explored**. This list is also called the **active list**.

3. Bounding: Compare the objective value of the current LP relaxation ($p^{*(rel)}_{curr}$) with the current best UB.

   If $p^{*(rel)}_{curr} \geq UB$ then **the current subproblem and all its children cannot provide a better solution than the one we already have found**.

   The current node can be **safely pruned**.

4. Updating the Global Lower Bounds and the Global Upper Bounds:

   The B&B method relies on the LB and the UB to prune subtrees and increase its efficiency. Every time it solves a subproblem (which is an LP) it generates a **LB** of the original IP. Also, if the optimal solution of the subproblem happens to satisfy the integrality constraints, it also generates an **UB**.

   The global lower bound is the minimum of the lower bounds of all active (unpruned) nodes.

   The global upper bound is the best (lowest) objective value of any integer-feasible solution found so far. It is updated whenever a new integer-feasible solution is discovered with a better objective value.

Analysis with the partitions

Let us denote by S^j the j-th partition of the feasible region of the original IP. It contains the original constraints ($Ax \leq b, x \in \mathbb{Z}^n$) and additional bound constraints on specific integer variables (e.g., $x_k \leq \alpha$ or $x_k \geq \beta$ for some $\alpha, \beta \in \mathbb{Z}$). Then the j-th subproblem (which is an IP) is defined as

$$\Phi(S^j) = \min_x \; c^T x \tag{5}$$

$$s.t. \; x \in S^j \tag{6}$$

where $\Phi(S^j)$ is its optimal solution.

We denote the optimal solution of the LP relaxation of the j-th subproblem by $\Phi_{LP}(S^j)$. Then we have

$$\Phi_{LP}(S^j) \leq \Phi(S^j) \leq \Phi_{UP}(S^j)$$

where $\Phi_{UP}(S^j)$ is the objective function value at any feasible point of the j-th subproblem.

An example of the B&B algorithm is shown in Figure 11. The original IP forms the root node of the B&B search tree. We can denote the feasible region of original IP as

$$S^0 = \{x \in \mathbb{Z}^2 | x_1 + x_2 \leq 6, x_1 \leq 4.5, x_2 \leq 3.5, x_1 \geq 0\}$$

Since its optimal solution is not integer, the feasible region is divided into two smaller regions

$$S^1 = \{x \in \mathbb{Z}^2 | x_1 + x_2 \leq 6, x_1 \leq 4.5, x_2 \leq 3.5, x_1 \geq 0, x_1 \leq \lfloor 2.5 \rfloor\}$$

$$S^2 = \{x \in \mathbb{Z}^2 | x_1 + x_2 \leq 6, x_1 \leq 4.5, x_2 \leq 3.5, x_1 \geq 0, x_1 \geq \lceil 2.5 \rceil\}$$

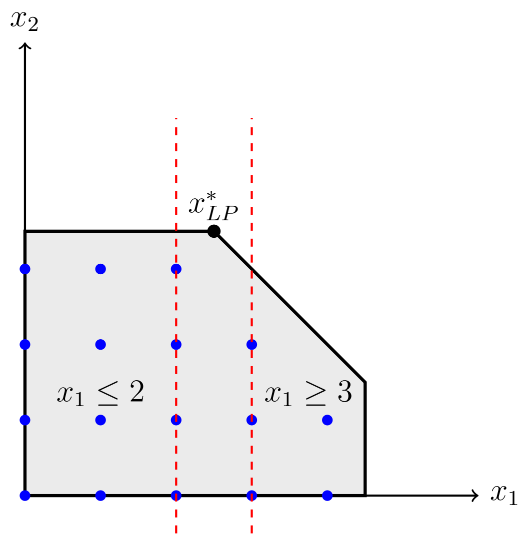

Figure 11: Branch and Bound main idea (divide and conquer)

Branch and Bound Algorithm

Consider the IP problem

$$\begin{aligned} \min_{x} \quad & c^T x \\ s.t. \quad & x \in P^{IP} \end{aligned}$$

where its feasible region is defined as

$$P^{(IP)} = \{x | Ax \leq b, x_i \in \mathbb{Z}, i \in \mathcal{I}\}$$

Iterations:

1. Branch: create/refine the partition of the P^{IP} and get S^j
2. Bound:
   - Compute lower and upper bounds:

     $$L_j = \Phi_{LB}(S^j) \quad \text{and} \quad U_j = \Phi_{UB}(S^j), \ \forall j$$

   - Update global bounds on $c^T x^*$

     $$L = \min_j L_j \quad \text{and} \quad U = \min_j U_j$$

3. If $U - L \leq \epsilon$ then the algorithm converged. Stop

Pruning strategy:
Assume U is non-increasing and L is non-decreasing.
If $L_j \leq \min_j U_j$, then S^j is active and must be considered.
Otherwise (else) it is inactive and can be pruned (it can be ignored).

Disjunctive constraints

Recall from lecture how we modeled the either-or constraint:

- **Goal**: either $a_1^T x \leq b_1$ or $a_2^T x \leq b_2$ is valid.
- We defined a binary auxiliary variable $y_1 \in \{0, 1\}$ and then we constructed the constraints:

$$\begin{aligned} & a_1^T x \leq b_i + y_1 M && (y_1 = 0 \text{ enforces } a_1^T x <= b_1) \\ & a_2^T x \leq b_2 + (1 - y_1) M && (y_1 = 1 \text{ enforces } a_2^T x <= b_s) \\ & y_1 \in \{0, 1\} \\ & x \in \mathbb{R}^n \end{aligned}$$

Suppose now we are given m "greater-than-or-equal ($\geq$)" constraints $a_i^T x \geq b_i, i = 1, \ldots, m$, where $a_i \in \mathbb{R}^n$ and $b_i \in \mathbb{R}$. How can we model the requirement: "at least k of the m constraints are satisfied"?

When we have m ($\geq$) constraints and we want at least k of them to be valid we

- define m binary variables $y_i \in \{0, 1\}, i = 1, \ldots, m$
- $y_i = 1$, if the i-th constraint is satisfied; $y_i = 0$, otherwise.
- Define the constraints

$$a_i^T x \geq b_i - (1 - y_i) M, \;\; i = 1, \ldots m$$

  They enforce the satisfaction of the constraints who can be valid (for those who are not valid we have $y_i = 0$ and $a_i^T x \geq b_i - M$ which makes $a_i^T x \geq b_i$ invalid, because $b_i - M < b_i$).

- The constraint

$$\sum_{i=1}^{m} y_i \geq k$$

  ensures that "at least k constraints are satisfied".

- Therefore, to model the requirement: "at least k of the m constraints are satisfied" we have to define m additional binary variables and add the constraints

$$a_i^T x \geq b_i - (1 - y_i) M, \;\; i = 1, \ldots m$$

$$\sum_{i=1}^{m} y_i \geq k$$

$$y_i \in \{0, 1\}, i = 1, 2, \ldots, m$$

Remark: if you had m "less-than-or-equal ($\leq$)" constraints $a_i^T x \leq b_i, i = 1, \ldots, m$, you should have used the inequalities

$$a_i^T x \leq b_i + (1 - y_i) M, \;\; i = 1, \ldots m$$

Production/Distribution problem

This example problem demonstrates how **effective** and **expressive** Integer Optimization is in **modeling complex real-life problems**.

A company produces a set of K products at I plants and ships these products to J market zones. We have the following data.

Data:

v_{ik}: cost of producing 1 unit of product k at plant i

c_{ijk}: cost of shipping 1 unit of product k from plant i to zone j

f_{ik}: fixed cost of producing product k at plant i

M_{ik}: maximal quantity of product k produced at plant i

m_{ik}: minimal quantity of product k that can be produced at plant i (that is, if plant i is selected to produce product k, it will produce at least m_{ik} units)

q_{ik}: production capacity of plant i for product k (that is, the maximum number of units of product k that plant i can produce).

Q_i: total capacity of plant i

d_{jk}: demand of product k at market zone j

Tasks:

1. Formulate the problem of minimizing the total cost as an IP problem

   Solution:

   First we have to define the decision variables:

   - $x_{ik} \in \mathbb{Z}^{I\times K}$: number of units of product k produced at plant i
   - $y_{ijk} \in \mathbb{Z}^{I\times J\times K}$: number of units of product k produced at plant i and shipped to market zone j
   - $z_{ik} \in \{0,1\}^{I\times K}$: binary variables indicating whether product k is produced at plant i

Then the IP problem can be defined as

$$\begin{aligned}
\min_{x,y,z} \quad & \sum_{i=1}^{I}\sum_{k=1}^{K} v_{ik}x_{ik} + \sum_{i=1}^{I}\sum_{k=1}^{K}\sum_{j=1}^{I} c_{ijk}y_{ijk} + \sum_{i=1}^{I}\sum_{k=1}^{K} f_{ik}z_{ik} \\
s.t. \quad & x_{ik} \leq M_{ik}z_{ik}, \ \ \forall i,k \\
& x_{ik} \geq m_{ik}z_{ik}, \ \ \forall i,k \\
& \sum_{k=1}^{K} q_{ik}x_{ik} \leq Q_i, \ \ \forall i \\
& d_{jk} = \sum_{i=1}^{I} y_{ijk}, \ \ \forall j,k \\
& \sum_{j=1}^{J} y_{ijk} = x_{ik}, \ \ \forall i,k \\
& x \in \mathbb{Z}^{I\times K}, y \in \mathbb{Z}^{I\times J\times K}, z \in \{0,1\}^{I\times K}
\end{aligned}$$

2. Express the following requirements as inequality constraints:

   (a) No plant can produce more than K_1 products
   Solution:
   $$\sum_{k=1}^{K} z_{ik} \leq K_1, \forall i$$

   (b) Every product can be produced in at most I_1 plants
   Solution:
   $$\sum_{i=1}^{I} z_{ik} \leq I_1, \forall k$$

   (c) For a particular product k_0, plant 3 must produce it if neither plant 1 nor plant 2 produces it
   Solution:
   $$z_{1k_0} + z_{2k_0} + z_{3k_0} \geq 1$$

   (d) Each market zone must be sourced by exactly one plant for all products
   Solution:
   Introduce new binary variables $w_{ijk} \in \{0,1\}^{I\times J\times K}$, such that
   $$w_{ijk} = \begin{cases} 1, \text{if product k is shipped to zone j from plant i} \\ 0, \text{otherwise} \end{cases}$$
   Note that $w_{ijk} = 0 \implies y_{ijk}$. Then we have the following constraints:
   $$0 \leq y_{ijk} \leq w_{ijk}M, \ \ \forall i,j,k$$
   $$\sum_{i=1}^{I} w_{ijk} \leq 1$$